The empirical recurrence for OEIS A208598, proved from an exact model

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Abstract

OEIS A208598 carries an empirical recurrence of order 4. It is true, and it is decidable rather than empirical. The entry's sequence is given exactly by a Burnside sum over the necklace group, from which a provably satisfies a MONIC linear recurrence of order $S = 21$, derived below and not assumed. The residual of the conjectured recurrence satisfies the same one, so whether it annihilates a is settled by a finite exact computation. No transfer matrix is involved and the count is not a walk.

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1 The conjecture

OEIS A208598 is "Number of 4-bead necklaces labeled with numbers -n..n not allowing reversal, with sum zero.". It has offset 1 and begins

$$6, 23, 60, 125, 226, 371, 568, 825, \ \ldots$$

The entry states, as a conjecture contributed by its author and never marked settled:

Empirical: a(n) = (4/3)*n^3 + 2*n^2 + (5/3)*n + 1.

As of the "Last modified" line on the live entry (Mar 18 2018 15:18:12, revision 12) this is still recorded as empirical, and nothing on the entry records it as proved.

2 The count is a Burnside sum

A necklace is an orbit of labellings under the rotation group, a bracelet an orbit under the dihedral group, so by Burnside's lemma $a(n)$ is the average over the group of the number of labellings a single element fixes. A labelling fixed by g is constant on each orbit of g, so writing x_j for the common value on the j-th orbit and w_j for its size, the fixed labellings with sum zero are the integer solutions of

$$\sum_j w_j x_j = 0, \qquad |x_j| \le n,$$

counted exactly by convolution. Summing over the group and dividing by its order gives $a(n)$.

3 The bound

Each term counts the integer points at height n of the cone $\{|x_j| \le t,\ \sum_j w_j x_j = 0\}$, whose rays are cut out by m of the defining hyperplanes; by Cramer the height of a primitive generator divides the determinant of their x-parts, which is 1 or one of the w_j. With at most $m+1$ rays in a simplicial cone, $\prod_{d|w_j \text{ some } j} \Phi_d(z)^{m+1}$ annihilates that term, and the product over the group's elements annihilates a. One more is added to the order, because the origin is a cell of that arrangement with no ray at all: its series is the constant 1, which over the common denominator has numerator degree equal to the denominator's rather than below it. So $S = 21$.

A rotation has all its orbits of one size, so its equation reduces to $\sum_j x_j = 0$ and every determinant is 1. A reflection of an odd bracelet fixes one bead and pairs the rest, giving weights 1 and 2, and forces $x_0 = -2(x_1 + \cdots)$ to be even; that term has period two, not one. An earlier version of this bound took S to be the number of beads for both, which is right for the necklaces and false for the bracelets — the fifth difference of A208826 is $-36, 48, -60, \ldots$ and never vanishes. The bound above is computed from the orbit sizes.

4 The proof

Lemma 1. *Let $A(z) = z^S - \sum_{j<S} \alpha_j z^j$ be the monic annihilator derived above and $q(z) = z^4 - \sum_i c_i z^{4-i}$ the characteristic polynomial of the conjectured recurrence. The residual $r(n) = a(n) - \sum_i c_i a(n-i)$ is a linear combination of shifts of a, so A annihilates r as well. Since $A(0) \neq 0$ the recurrence it gives runs in both directions, and S consecutive zeros of r force r to vanish at every later index.*

Proof. A annihilates a, hence every shift of a, hence any linear combination of shifts; that is r. A monic recurrence with nonzero constant term determines each term from the S before it and each from the S after it, so a run of S zeros propagates. □

Theorem 2.

$$a(n) \;=\; 4\,a(n-1) - 6\,a(n-2) + 4\,a(n-3) - a(n-4)$$

for every $n > 0$.

Proof. The terms $a(0), \ldots$ were computed exactly in integer arithmetic from the model, extended by A where the model itself was not evaluated, and $r(n)$ formed for each index. Those integers vanish from $n = 0 + 1$ onwards, and the run exceeds $S + 4$, which by Lemma 1 settles every larger index at once. □

5 Verification

Three checks, each independent of the algebra above.

First, the model reproduces all 40 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry: it is built by reading the entry's English, and a misreading gives different counts.

Second, the conjectured recurrence was evaluated directly on those published terms, with no model involved, and holds wherever the proved range and the data overlap.

Third, the derived annihilator A was itself tested on terms it had not been given: the model was evaluated beyond the S values that determine it and A reproduced them. A bound that is too small fails this test, which is why it is run.

References

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